

TR

NFSU × IBM · BOB AI HACKATHON · TRACK 1 · PS-03

Evidence Triage Console

AI-assisted crime scene evidence prioritization

Bob extracts & classifies. **Deterministic code scores, flags & ranks.**

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THE PROBLEM

Too much evidence, no defensible way to triage it

2019 Hyderabad homicide. Investigators faced thousands of photographs and hundreds of items. A DNA-bearing cigarette butt — the single most identifying piece — was nearly missed on the first sweep, while overloaded labs delayed testing for weeks.

3,000+

photographs at one scene

200+

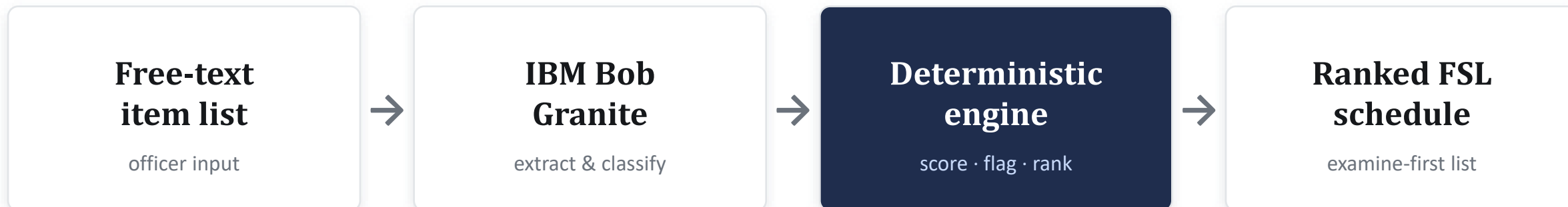
physical items recovered

Weeks

of FSL testing backlog

The gap is not evidence — it is deciding what the lab examines first, in a way that holds up in court.

The AI never decides the priority



The rule that makes it court-admissible

*Bob does only the language step — it classifies items into a fixed taxonomy. **It never assigns a score, never ranks, never decides urgency.** Every priority is computed by pure Python from a versioned, hashed ruleset — so the same input always yields the same schedule.*

Four stages, one trust boundary

1 Extract

IBM Bob / Granite reads the free text and returns structured, classified items. Temperature 0, strict validation, hallucinations rejected and routed to manual review — never guessed.

2 Score

Pure Python: base weight × crime-type multiplier + degradation risk + context-flag bonus, clamped 0–100 with Decimal / round-half-up. No model, no randomness.

3 Flag urgency

An independent boolean: score ≥ 30 OR a force-urgent flag (electronic, weather-exposed, biological). A low-scoring but perishable item is never buried.

4 Rank & schedule

Stable sort — urgency, then score, then item id — into a ranked FSL examination schedule, each item stamped with the ruleset's SHA-256 hash.

A ranked, explainable examination schedule

Homicide scene · reference case (8 items, 5 shown) · generated deterministically

#	ITEM & CATEGORY	STATUS	SCORE
1	Bloodstained kitchen knife Biological / DNA	URGENT	25.5
2	Locked smartphone, powered on Digital / Electronic	URGENT	19.6
3	Single strand of hair on the victim's collar Biological / DNA	URGENT	18.5
5	Cigarette butt near the gate ← nearly missed in the real case Biological / DNA	URGENT	15.5
7	Empty liquor bottle, fingerprints Pattern / Impression	ROUTINE	8.6

Load-bearing, not name-dropped

An investigator talks to Bob. Bob calls two MCP tools — and the boundary between them enforces our design rule.

extract_evidence

LANGUAGE · watsonx Granite

Reads free text.

Returns structured, classified items in a fixed taxonomy — the part LLMs are good at.

score_evidence

DETERMINISTIC · PURE PYTHON

Scores and ranks.

Bob must call this to get a priority — it cannot compute the score itself. Same input, same output, every time.

The tool boundary is the guarantee: the language model structures the data; the deterministic tool decides the priority.

WHY IT CAN BE TRUSTED

Reproducible and auditable by construction

57

tests green

Zero-tolerance exact-match tests
pin the scoring formula.

1000×

identical

The scorer runs a thousand times
on one input — byte-identical every
run.

SHA-256

on every result

Each schedule is stamped with the
hash of the exact ruleset used.

Replay

any past run

A SQLite audit log reproduces old
results without calling the model.

Python 3.11+ · FastAPI · Pydantic v2 (strict) · watsonx.ai Granite · MCP · SQLite · pytest

From an overwhelming pile to a defensible worklist

- For the officer at 2 a.m.: a ranked, explainable list of what to seize and rush first.
- For the lab: the highest-value, most-perishable evidence tested before the backlog buries it.
- Next: calibrate weights against real FSL outcomes; e-FSL submission; multilingual intake.

Bob does the language. The rules decide the priority. And we can prove it.